



MID TERM EVEN SEMESTER EXAMINATION-2026

Roll no.....

Name of the Program & Semester: B.Tech II SEM.

Name of the Course: Programming in C

Course Code: TCS202

Time: 1.5 hours

Maximum Marks: 50

Note:

- (i) Answer all the questions except Que(5) by choosing any one of the sub-questions.
- (ii) All 5 sub-questions in Que(5) are compulsory.
- (iii) Each question carries 10 marks.

Que(1)	(10 marks)				
a)	Explain the structure of a C program. Discuss the following terms: 1. break and continue 2. precedence and associativity 3. qualifiers 4. typecasting	CO1			
OR					
b)	Explain the Logical, Increment/Decrement, Bitwise and Ternary operators in C. Write a C program to find the roots of a quadratic equation for the discriminant equal to zero and greater than zero.				
Que(2)	(10 marks)				
a)	Draw a flowchart to print following pattern. 1 0 1 1 0 1 0 1 0 1 1 0 1 0 1	CO2			
OR					
b)	Draw a flowchart to read day number from the user and print the next day of the week. Implement a C program for the same using a switch-case statement. Assume: 1-Sunday, 2-Monday, 3-Tuesday, 4-Wednesday, 4-Thursday, 5-Friday, 6-Friday, 7-Saturday. <table border="1"><tr><td>Sample Input:</td><td>Sample Output:</td></tr><tr><td>Inputted day number: 3</td><td>Output next day: Wednesday</td></tr></table>		Sample Input:	Sample Output:	Inputted day number: 3
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Inputted day number: 3	Output next day: Wednesday				
Q3.	(10 marks)				
a)	Explain the concept of recursion in C. How it is different from iteration? Discuss at least three points. Develop a recursive function in C to find the sum of following series. 1 - 3 + 5 - 7 + 9 N (where N is odd)	CO4			
OR					
b)	Describe the concept of Storage Classes in C. Discuss in detail with syntax and examples, also compare them using Storage Area, Initial value, Scope and Lifetime.				



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Q4.	(10 marks)		
a)	Develop a C code which passes a positive integer number into a function <i>check()</i> . If the number is divisible by 3 then the function returns its sum of digit and if not, then the function returns its multiplication of digits.		CO4
OR			
b)	Develop a C code which passes an integer number into a function <i>strong()</i> and checks whether the number is a strong number or not. Create a separate function <i>fact()</i> to calculate factorial. Note: Strong number is a number whose sum of factorial of each digit is equivalent to that number. For eg. 145(1! + 4! + 5!)		
Q5.	Predict the output of the following code and justify your answer. Assume below code snippets are free from syntax errors. (10 marks)		
i) #include<stdio.h> int main() { int a = 5, b = 3; printf("%d ", a++ + ++b); printf("%d %d", a, b); return 0; }		ii) #include<stdio.h> int main() { int x = 10, y = 5, z; z = x > y > 2; printf("%d", z); return 0; }	CO1 CO3
iii) #include<stdio.h> void Disp(int x) { static int X=5; printf("%d %d \n", x, X); --X; } int main() { int i; static int x=4; for(i=0;i<3;i++) Disp(x++); return 0; }		iv) #include<stdio.h> int main() { int n = 2; switch(n) { case 1: printf("One "); case 2: printf("Two "); case 3: printf("Three "); default: printf("Default"); } return 0; }	
v) #include<stdio.h> int Mod(int x, int y) { if(y == 0) return x; return(Mod(y, x%y)); } int main() { printf("%d", Mod(10, 35)); return 0; }			